

Unequal-Twist Stieltjes Jets

Monodromy Obstructions, Convergent Harmonic Expansions,
Gamma Convolutions, and Multishift Zeta Finite Parts

Research continuation prepared for Vladimir Reshetnikov
ProveIt programme • Prepared with ChatGPT

10 October 2026

Abstract

The current ProveIt manuscript proves finite partial-fraction formulas for integer-order kernels with unequal frequency shifts, but leaves mixed Stieltjes order jets in a prescribed finite Hurwitz basket as a separate question. We settle a precise functional version of that question. A product of single-shift logarithmic multipliers belongs to the finite single-shift integer-order jet span exactly in the rational case or when only one shift is active. The obstruction is analytic monodromy, not a failed numerical relation search and not a statement about individual period values.

The positive counterpart is a normally convergent, all-order expansion of every finite mixed-twist convolution into one central twist. Its coefficients are finite Pochhammer derivatives, hence generalized harmonic and Stirling polynomials. Finite Fourier corrections remove the restriction that the twists lie in one frequency interval. A symmetric specialization gives explicit Stieltjes contact terms and ordinary convolutions of rational-grid log-Gamma functions. A unilateral version yields exact multishift Lerch identities, all ordinary primitives, a meromorphic zeta continuation, and an all-index formula for logarithmic cutoff constants in single Hurwitz-zeta derivatives. These constants satisfy a lower-degree divided-difference recursion and a Gamma antiderivative identity. At the zero-order spectral intersection, we compute every ray jet: the value is direction-dependent, whereas the first derivative is a finite additive log-Gamma expression.

Full proofs, 824 finite exact assertions, and independent numerical checks accompany the formulas. Classical mechanisms are credited; no claim of global priority, proof-assistant formalization, or resolution of fixed-value S_6/S_8 identities is made.

Source anchor. Targeted source reading is pinned to commit
`b1df802799851c62a860570c3ac78ee31b9f5c04`.

The complete repository and the binary incoming archives were not exhaustively audited. The package is additive and makes no repository changes.

Contents

1	The question and the exact scope of the answer	3
2	Conventions and the normalized twisted family	3
3	A convergent expansion for every finite set of twists	4
3.1	Arbitrary real nonintegral twists and finite mode corrections	5
4	All mixed order jets and their harmonic coefficients	6
4.1	A symmetric first-jet identity with its contact term	6
5	A complete finite-closure classification in the integer ladder	7
6	Covariant primitives and ordinary log-Gamma convolutions	9
6.1	Finite rational-grid Gamma representatives	9
6.2	Two half-grid zeta-series evaluations	10
7	The unilateral multishift Lerch family and its primitives	11
7.1	Differential relations and all ordinary primitives	12
8	Logarithmic cutoff constants: all indices and exact recursions	12
8.1	A Gamma antiderivative and polygamma collision jets	14
9	Spectral rays at the zero-order intersection	14
10	Cyclotomic coordinates and explicit scalar instances	15
11	Source audit, reproducibility, and integration boundaries	16
11.1	What was actually inspected	16
11.2	Mathematical safeguards for integration	16
11.3	Executed checks and their evidentiary role	17
11.4	Suggested placement	17
12	Further research questions	18
	Conclusion	19

1 The question and the exact scope of the answer

The canonical unequal-twist chapter defines K_α^θ by the Fourier multiplier $(2\pi i(n + \theta))^{-\alpha}$, proves a Dirichlet beta mixture for arbitrary orders in a common frequency interval, and proves finite partial fractions at positive integer orders [1]. The preceding twisted chapter fixes the meromorphic Lerch family, its endpoint distributions, its Gamma normalization, and its rational Hurwitz coordinates [2]. We retain those conventions. The editorial ledger explicitly separates these established results from finite reduction of mixed Stieltjes jets [3].

There are three distinct questions. A *functional identity* in a finite span of single-twist integer-order jets is a statement about all Fourier coefficients. A *convergent single-twist expansion* can use unbounded positive orders. A *special-value identity* is an equality at a particular argument and may hold without any functional reduction. Our nonclosure theorem concerns the first question; our expansion answers the second. It neither proves nor disproves the third.

The central contributions are Theorem 3.1, its all-jet form (4.1), the finite-closure classification Theorem 5.3, the ordinary Gamma convolution (6.8), the cutoff formula (8.2), and the spectral-ray formula (9.3). They form one chain rather than unrelated examples: all arise by resolving several shifted denominators about one centre, keeping the logarithmic decorations before taking limits.

The binomial theorem, partial fractions, Fourier uniqueness, the Hurwitz shift calculus, Gamma products, and cyclotomic filters are classical. DLMF supplies the relevant normalizations [4, 5, 6, 7, 8]. The broader analytic and multivalued setting of Lerch functions is treated by Lagarias and Li [9]. We do not claim that a literature priority search is complete. In particular, a derived special case of a classical Gamma product is identified as such, not advertised as a new special-function discovery.

2 Conventions and the normalized twisted family

For $0 < \theta < 1$ and $n \in \mathbb{Z}$, set

$$\lambda_{n,\theta} = 2\pi i(n + \theta), \quad L_{n,\theta} = \log(2\pi|n + \theta|) + \frac{i\pi}{2} \operatorname{sgn}(n + \theta). \quad (2.1)$$

All powers of λ use this logarithm. Fourier coefficients are $\hat{f}(n) = \int_0^1 f(x)e^{-2\pi i n x} dx$, and circular convolution multiplies Fourier coefficients. Define

$$\widehat{K}_\alpha^\theta(n) = e^{-\alpha L_{n,\theta}}, \quad J_m^{r,\theta} = (-1)^m \partial_\alpha^m K_\alpha^\theta \Big|_{\alpha=r}. \quad (2.2)$$

Thus $\widehat{J}_m^{r,\theta}(n) = L_{n,\theta}^m \lambda_{n,\theta}^{-r}$. The order r may be complex until an integer assumption is stated. In particular, $J_0^{0,\theta} = K_0^\theta = \delta_0$; the zero Fourier coefficient is retained, and is not a zero *shifted* frequency.

On a compact set of orders the multipliers and any fixed number of order derivatives grow at most polynomially in $|n|$. Pairing with a smooth periodic test function therefore proves that K_α^θ is entire as a distribution-valued function. Products of these polynomially bounded sequences again define distributions, so all convolutions used below are well-defined. This is not pointwise multiplication of singular functions.

Write $W_s^\theta = K_{1-s}^\theta$ and $Z_s^\theta = \Gamma(1-s)W_s^\theta$. The canonical normalization is

$$Z_{1+u}^\theta = -\frac{\delta_0}{u} + \sum_{m \geq 0} \frac{(-1)^m}{m!} U_m^\theta u^m. \quad (2.3)$$

Away from $x = 0$, the family has the ordinary representative

$$Z_s^\theta(x) = e^{-2\pi i \theta x} \Phi(e^{-2\pi i \theta}, s, x), \quad 0 < x < 1, \quad (2.4)$$

with the endpoint continuation fixed by (2.3); see [2]. Expanding $\Gamma(-u)$ gives the useful dictionary

$$E_m^\theta := J_m^{0,\theta}, \quad (2.5)$$

$$E_1^\theta = -U_0^\theta - \gamma \delta_0, \quad (2.5)$$

$$E_2^\theta = 2U_1^\theta + 2\gamma U_0^\theta + (\gamma^2 - \zeta(2))\delta_0. \quad (2.6)$$

Here γ is Euler's constant. In particular the Gamma normalization cannot be suppressed without changing point-supported terms.

For scalar functions we use

$$\zeta(1+u, a) = \frac{1}{u} + \sum_{m \geq 0} \frac{(-1)^m}{m!} \gamma_m(a) u^m, \quad \gamma_0(a) = -\psi(a), \quad a > 0. \quad (2.7)$$

A superscript (j) on ζ or Φ denotes differentiation in the *order*, not in the final parameter. Harmonic numbers are $H_N^{(j)} = \sum_{\ell=1}^N \ell^{-j}$, with $H_0^{(j)} = 0$ and $H_N = H_N^{(1)}$. The unsigned Stirling number of the first kind is $[k]_q$, with $[0]_q = 1$ and value zero outside $0 \leq q \leq k$.

3 A convergent expansion for every finite set of twists

Let $\theta_1, \dots, \theta_d \in (0, 1)$, choose $c \in (0, 1)$, and put

$$d_j = \theta_j - c, \quad R = \max_j |d_j|, \quad \eta = \min(c, 1 - c), \quad A = \sum_{j=1}^d \alpha_j. \quad (3.1)$$

Assume $R < \eta$. Such a centre always exists: the midpoint of the least and greatest twist works. Define the finite polynomials

$$h_k(\alpha; \mathbf{d}) = \sum_{k_1 + \dots + k_d = k} \prod_{j=1}^d \frac{(\alpha_j)_{k_j}}{k_j!} d_j^{k_j}, \quad (3.2)$$

$$\sum_{k \geq 0} h_k t^k = \prod_{j=1}^d (1 - d_j t)^{-\alpha_j}, \quad (3.3)$$

$$h_0 = 1, \quad k h_k = \sum_{\ell=1}^k \left(\sum_j \alpha_j d_j^\ell \right) h_{k-\ell}. \quad (3.4)$$

The recurrence follows by logarithmic differentiation of (3.3), and supplies an exact coefficient algorithm.

Theorem 3.1 (Entire multicentre expansion). *For all $\alpha \in \mathbb{C}^d$,*

$$K_{\alpha_1}^{\theta_1} * \dots * K_{\alpha_d}^{\theta_d} = \sum_{k=0}^{\infty} (-2\pi i)^k h_k(\alpha; \mathbf{d}) K_{A+k}^c. \quad (3.5)$$

The series converges locally uniformly in the distribution topology, may be differentiated any finite number of times in the orders, and retains every Fourier coefficient, including $n = 0$.

Proof. At a fixed mode all $n + \theta_j$ have the same sign as $n + c$. Since $|d_j/(n + c)| < 1$, the real number $1 + d_j/(n + c)$ is positive. Hence

$$L_{n,\theta_j} = L_{n,c} + \log \left(1 + \frac{d_j}{n + c} \right)$$

with no missing phase. The ordinary binomial theorem gives

$$\prod_j \lambda_{n,\theta_j}^{-\alpha_j} = \lambda_{n,c}^{-A} \sum_{k \geq 0} \frac{(-1)^k h_k(\boldsymbol{\alpha}; \mathbf{d})}{(n + c)^k}. \quad (3.6)$$

This is precisely the Fourier coefficient of (3.5).

For normal convergence, choose R_0 strictly between R and η . The generating function (3.3) is bounded on $|t| = R_0^{-1}$ uniformly on a compact set of $\boldsymbol{\alpha}$; Cauchy's estimate gives $|h_k| \leq CR_0^k$. The sum of the absolute values of the terms in (3.6) is therefore bounded by a constant times $|\lambda_{n,c}^{-A}|$, uniformly in n . This last factor has at most polynomial growth uniformly on the compact order set. Multiply by the rapidly decreasing Fourier coefficients of a test function and sum. This proves absolute double-sum convergence and normal convergence of all pairings. Cauchy's formula in the order variables justifies all order derivatives. Fourier uniqueness proves the identity. $\square$

Remark 3.2 (What the expansion accomplishes). The series is geometric in the shift displacement, not merely asymptotic. Its coefficients are finite polynomials, while its single-twist orders increase without bound. The result is compatible with finite partial fractions at integer orders, but it is not obtained by differentiating an identity stated only on the integer lattice.

3.1 Arbitrary real nonintegral twists and finite mode corrections

The common-interval hypothesis can be removed from the expansion, though not from the unsubtracted beta mixture used in the manuscript. Let now $\theta_j \in \mathbb{R} \setminus \mathbb{Z}$ be arbitrary and choose any $c \in \mathbb{R} \setminus \mathbb{Z}$. Take an integer N large enough that

$$|n + c| > R := \max_j |\theta_j - c| \quad (|n| \geq N). \quad (3.7)$$

For a distribution F , let $P_N F$ retain the modes $|n| < N$, and put $K_\alpha^{c,[N]} = (I - P_N)K_\alpha^c$.

Corollary 3.3 (Exact finite-mode repair). *With h_k as in (3.2),*

$$\begin{aligned} K_{\alpha_1}^{\theta_1} * \cdots * K_{\alpha_d}^{\theta_d} &= \sum_{|n| < N} e^{2\pi i n x} \prod_j \lambda_{n,\theta_j}^{-\alpha_j} \\ &\quad + \sum_{k \geq 0} (-2\pi i)^k h_k(\boldsymbol{\alpha}; \mathbf{d}) K_{A+k}^{c,[N]}. \end{aligned} \quad (3.8)$$

The second series is normally convergent in distributions and can be differentiated in every order parameter.

Proof. Condition (3.7) guarantees both the common sign and the convergent binomial expansion at all retained tail modes. Their denominators are bounded below by a number strictly greater than R . The proof of Theorem 3.1 applies there. The exceptional modes are restored with their actual logarithms, so no incorrect common phase is assigned to them. $\square$

4 All mixed order jets and their harmonic coefficients

For multi-indices, write $|\mathbf{m}| = \sum_j m_j$, $\binom{\mathbf{m}}{\mathbf{q}} = \prod_j \binom{m_j}{q_j}$, and $\partial^{\mathbf{q}} = \prod_j \partial_{\alpha_j}^{q_j}$. Put $R_* := \sum_j r_j$ and $M := |\mathbf{m}|$; R_* is an order sum, not the geometric radius R of (3.1).

Theorem 4.1 (The all-jet coefficient formula). *For nonnegative integers m_j and arbitrary complex r_j ,*

$$J_{m_1}^{r_1, \theta_1} * \dots * J_{m_d}^{r_d, \theta_d} = \sum_{k \geq 0} (-2\pi i)^k \sum_{\mathbf{q} \leq \mathbf{m}} (-1)^{|\mathbf{q}|} \binom{\mathbf{m}}{\mathbf{q}} (\partial^{\mathbf{q}} h_k)(\mathbf{r}; \mathbf{d}) J_{M-|\mathbf{q}|}^{R_*+k, c}. \quad (4.1)$$

The same formula applies to (3.8), with the differentiated finite head and the corresponding projected central jets.

Proof. Apply $(-1)^M \partial^{\mathbf{m}}$ to (3.5). In each Leibniz term, $|\mathbf{q}|$ derivatives act on h_k and the remaining $M - |\mathbf{q}|$ derivatives act on K_{A+k}^c . The latter give $(-1)^{M-|\mathbf{q}|} J_{M-|\mathbf{q}|}^{R_*+k, c}$; the resulting sign is $(-1)^{|\mathbf{q}|}$. The normal convergence already proved justifies the operation, including at zero and negative integer orders. $\square$

Every coefficient in (4.1) is a finite sum of products of

$$D_q(r, k) = \partial_u^q \frac{(u)_k}{k!} \Big|_{u=r}. \quad (4.2)$$

For positive integer r , let $\mathcal{H}_j = H_{r+k-1}^{(j)} - H_{r-1}^{(j)}$. The exponential complete Bell polynomial B_q is specified by $\exp(\sum_{j \geq 1} x_j t^j / j!) = \sum_{q \geq 0} B_q(x_1, \dots, x_q) t^q / q!$. Then

$$D_q(r, k) = \frac{(r)_k}{k!} B_q(\mathcal{H}_1, -1! \mathcal{H}_2, 2! \mathcal{H}_3, \dots, (-1)^{q-1} (q-1)! \mathcal{H}_q). \quad (4.3)$$

This follows by taking logarithms of $(r+u)_k / (r)_k = \prod_{j=0}^{k-1} (1 + u/(r+j))$. At the important zero order one must not divide by $(0)_k$:

$$D_q(0, k) = \frac{q!}{k!} \left[\frac{k}{q} \right], \quad k, q \geq 0. \quad (4.4)$$

In particular, for $k \geq 1$,

$$\begin{aligned} D_1(0, k) &= \frac{1}{k}, & D_2(0, k) &= \frac{2H_{k-1}}{k}, \\ D_3(0, k) &= \frac{3(H_{k-1}^2 - H_{k-1}^{(2)})}{k}. \end{aligned} \quad (4.5)$$

These identities also follow from $(u)_k / k! = u \prod_{j=1}^{k-1} (1 + u/j) / k$. They explain why differentiating before setting an order to zero is essential: the value $(0)_k = 0$ does not imply that its jets vanish.

4.1 A symmetric first-jet identity with its contact term

Set $a = c - d$, $b = c + d$, with $|d| < \min(c, 1 - c)$, and define

$$\eta_k = H_{2k-1} - H_{k-1}, \quad k \geq 1. \quad (4.6)$$

Corollary 4.2 (Symmetric mixed Stieltjes generators). *The following identity holds in periodic distributions:*

$$E_1^a * E_1^b = E_2^c - \sum_{k \geq 1} \frac{(2\pi i d)^{2k}}{k} \left(J_1^{2k,c} + \eta_k K_{2k}^c \right). \quad (4.7)$$

In the canonical unnormalized Stieltjes coordinates this is

$$\begin{aligned} U_0^a * U_0^b &= 2U_1^c + 2\gamma U_0^c - \gamma(U_0^a + U_0^b) - \zeta(2)\delta_0 \\ &\quad - \sum_{k \geq 1} \frac{(2\pi i d)^{2k}}{k} \left(J_1^{2k,c} + \eta_k K_{2k}^c \right). \end{aligned} \quad (4.8)$$

Proof. At a mode put $L = L_{n,c}$ and $u = d/(n+c)$. Then

$$L_{n,a} L_{n,b} = L^2 + L \log(1-u^2) + \log(1-u) \log(1+u).$$

The elementary power-series identity

$$\log(1-u) \log(1+u) = - \sum_{k \geq 1} \frac{\eta_k}{k} u^{2k} \quad (4.9)$$

follows from $\log^2(1-u) = \sum_{j \geq 2} 2H_{j-1} u^j / j$ and $2 \log(1-u) \log(1+u) = \log^2(1-u^2) - \log^2(1-u) - \log^2(1+u)$. This proves (4.7) mode by mode. Substituting (2.5)–(2.6) gives (4.8). The first correction term in (4.7) is $-(2\pi i d)^2 (J_1^{2,c} + K_2^c)$; this checks both signs. $\square$

When $d = 0$, (4.8) becomes $U_0^c * U_0^c = 2U_1^c - \zeta(2)\delta_0$, in agreement with the one-twist Gamma normalization. Omitting the point mass gives an incorrect identity even at the zero Fourier coefficient.

5 A complete finite-closure classification in the integer ladder

The next theorem concerns a *specified function space*. Coefficients in a linear combination are constants with respect to the frequency variable; they may depend on the fixed shifts. A pointwise evaluation of the resulting functions is a different problem.

Definition 5.1. Let $\mathcal{V}$ consist of all finite linear combinations of sequences

$$(n+c)^{-j} \log^q(n+c), \quad j \in \mathbb{Z}, \quad q \in \mathbb{Z}_{\geq 0}, \quad (5.1)$$

where finitely many centres c are allowed and all expressions are understood for sufficiently large positive integers n on their standard large-positive branch. Equality is allowed to fail at finitely many n . The centres need not be prescribed in advance.

The allowance $j \leq 0$ includes polynomials and hence the Fourier multipliers of point masses and their derivatives at the origin. The factors $2\pi i$ and the additive logarithm constant in (2.1) do not change this span: a finite binomial change of logarithmic coordinates absorbs them.

Lemma 5.2 (From eventual sequence equality to a function identity). *Suppose $F(y)$ is a finite sum of products of integer powers of $y+c$ and nonnegative integer powers of $\log(y+c)$. If $F(n) = 0$ for every sufficiently large positive integer, its analytic germ at large positive y is identically zero.*

Proof. For sufficiently large $|y|$ in a fixed sector about the positive real axis, expand each shifted power and write $\log(y + c) = \log y + \log(1 + c/y)$. Thus

$$F(y) = \sum_{p \geq p_0} y^{-p} P_p(\log y),$$

where the coefficients of each power of $\log y$ form convergent Laurent series in $1/y$, and the degrees of all P_p have a common finite bound D . If not all P_p vanish, choose the least p with $P_p \neq 0$. At positive integers,

$$F(n) = n^{-p} \{P_p(\log n) + O(n^{-1}(\log n)^D)\}.$$

A nonzero polynomial in $\log n$ dominates the remainder for all sufficiently large n . This contradicts eventual vanishing. Therefore all Laurent coefficients vanish and the analytic germ is zero. $\square$

Theorem 5.3 (Finite single-centre closure criterion). *Let $a_1, \dots, a_d$ be distinct real shifts, and let r_j, m_j be nonnegative integers. Remove inactive sites with $r_j = m_j = 0$. Then the sequence*

$$F(n) = \prod_{j=1}^d (n + a_j)^{-r_j} \log^{m_j}(n + a_j) \quad (5.2)$$

belongs to $\mathcal{V}$ if and only if either all $m_j = 0$, or there is at most one active site.

Proof. If all $m_j = 0$, ordinary partial fractions give the required finite representation. With one active site the target is itself a generator.

Conversely, an eventual sequence representation gives an analytic function identity by Lemma 5.2. Continue this identity along loops around the finitely many points $-a_j$ and the candidate centres. Let Δ_a denote continuation once counterclockwise around $-a$ minus the original germ. It replaces $\log(y + a)$ by $\log(y + a) + 2\pi i$, and fixes logarithms at every other centre and all integer rational powers. These operations commute on the functions under consideration.

Suppose two sites, say a and b , have positive logarithmic degree. Then $\Delta_a \Delta_b F$ is nonzero: it is the nonzero rational prefactor times the product of the two nonzero finite differences of positive-degree logarithm powers and the other factors. Every generator in (5.1), however, is killed by $\Delta_a \Delta_b$. This contradicts the continued identity.

It remains to consider exactly one logarithmically decorated site a , of degree $m \geq 1$, and another active site b with $r_b > 0$. Write $F(y) = R(y) \log^m(y + a)$, where R is rational and has a pole at $-b$. Group the putative single-centre representation by centre and log degree. Its a -centred portion is $\sum_q R_q(y) \log^q(y + a)$, with each R_q a Laurent polynomial in $y + a$. All other portions have trivial monodromy around $-a$. Because $\Delta_a^{m+1} F = 0$, this identity forces $R_q = 0$ for $q > m$. Indeed, a polynomial in $\log(y + a)$ with Laurent-polynomial coefficients has a unique such expansion, by the same leading-power argument near $y = -a$ as in Lemma 5.2; its highest log degree cannot cancel a lower one under repeated finite differences. Applying Δ_a^m now yields

$$(2\pi i)^m m! R(y) = (2\pi i)^m m! R_m(y).$$

The right side can have no finite pole except at $-a$, whereas the left has a pole at $-b$. This is impossible. $\square$

Corollary 5.4 (Mixed twisted jets do not generally close finitely). *For distinct nonintegral twists, a convolution of jets $J_{m_j}^{r_j, \theta_j}$ with nonnegative integer orders satisfies the same criterion for membership in any finite constant-coefficient span of single-twist integer-order jets. Finite Fourier corrections and point masses or their derivatives at the origin cannot repair the obstruction. In particular $E_1^a * E_1^b$, and also $U_0^a * U_0^b$, do not admit such a finite reduction when $a \neq b$.*

Proof. Apply the theorem to positive Fourier modes. Scaling the powers by $2\pi i$ and translating the logarithms by $\log(2\pi) + i\pi/2$ are finite triangular transformations. For the U_0 product, (2.5) shows that its difference from $E_1^a * E_1^b$ lies in the single-twist span. Equality modulo finite Fourier support is already covered by eventual sequence equality. $\square$

Limits of the conclusion. This is a functional nonclosure result for the integer-order ladder, not a proof of numerical period independence, not non-elementarity at a fixed x , and not a theorem excluding arbitrary argument-dependent coefficients, arbitrary complex-order bases, or every possible special-function enlargement. In particular, evaluating at a rational x can introduce cancellations which the theorem does not address. The infinite expansion (4.1) is not in conflict with nonclosure: it uses infinitely many integer orders.

6 Covariant primitives and ordinary log-Gamma convolutions

For $D_c = \partial_x + 2\pi ic$, the nonintegrality of c makes D_c invertible on periodic distributions. Its inverse of order $p \geq 0$ has multiplier $\lambda_{n,c}^{-p}$ and hence

$$D_c^{-p} J_m^{r,c} = J_m^{r+p,c}. \quad (6.1)$$

It may be applied termwise to (4.1) or (4.7). These are *covariant* primitives, not an unqualified inverse of the ordinary periodic derivative. If $D_c u = f$, then $(e^{2\pi icx} u)' = e^{2\pi icx} f$, so the integrating factor converts them into ordinary weighted antiderivatives.

For positive integer r , the ordinary representative of a central kernel is

$$K_r^c(x) = \frac{e^{-2\pi icx}}{\Gamma(r)} \Phi(e^{-2\pi ic}, 1-r, x), \quad 0 < x < 1. \quad (6.2)$$

Differentiating the entire order identity gives

$$J_1^{r,c}(x) = \frac{e^{-2\pi icx}}{\Gamma(r)} \left(\Phi^{(1)} + \psi(r)\Phi \right), \quad (6.3)$$

$$J_2^{r,c}(x) = \frac{e^{-2\pi icx}}{\Gamma(r)} \left(\Phi^{(2)} + 2\psi(r)\Phi^{(1)} + (\psi(r)^2 - \psi'(r))\Phi \right), \quad (6.4)$$

where every Φ on the right is evaluated at $(e^{-2\pi ic}, 1-r, x)$. At integer r , the coefficients reduce to $\psi(r) = H_{r-1} - \gamma$ and $\psi'(r) = \zeta(2) - H_{r-1}^{(2)}$. All higher jets follow by the same Leibniz rule for $1/\Gamma(r)$; the general master derivative avoids any division by $\Gamma(0)$ at the exceptional order zero.

6.1 Finite rational-grid Gamma representatives

Let $\theta = p/q \in (0, 1)$ be rational and $z = e^{-2\pi ip/q}$. Splitting an ordinary Lerch series into residues gives

$$\Phi(z, s, x) = q^{-s} \sum_{\ell=0}^{q-1} z^\ell \zeta\left(s, \frac{x+\ell}{q}\right). \quad (6.5)$$

This extends to all s ; the poles of the individual Hurwitz functions cancel because $\sum z^\ell = 0$. Using $\zeta'(0, y) = \log \Gamma(y) - \frac{1}{2} \log(2\pi)$ and $\Phi(z, 0, x) = (1-z)^{-1}$ gives an explicit representative for $P_\theta := J_1^{1,\theta}$:

$$P_{p/q}(x) = e^{-2\pi ipx/q} \left\{ \sum_{\ell=0}^{q-1} z^\ell \log \Gamma\left(\frac{x+\ell}{q}\right) - \frac{\gamma + \log q}{1-z} \right\}. \quad (6.6)$$

The real positive branch of $\log \Gamma$ is used on these arguments. The $\log q$ and Euler-constant term are necessary, not optional normalization choices.

The Fourier coefficients of P_θ are $L_{n,\theta}/\lambda_{n,\theta}$, which are square summable. Thus $P_\theta \in L^2(\mathbb{R}/\mathbb{Z})$, and the circular convolution of any two such functions is an ordinary continuous function. The integral is not a finite part.

Theorem 6.1 (An ordinary mixed Gamma convolution). *Let $a = c - d$, $b = c + d$, with $|d| < \min(c, 1 - c)$. Set*

$$S_0 = 0, \quad S_k = \sum_{\ell=1}^k \frac{\eta_\ell}{\ell}. \quad (6.7)$$

Then, for every $x \in \mathbb{R}/\mathbb{Z}$,

$$\int_0^1 P_a(t) P_b(\{x - t\}) dt = \sum_{k=0}^{\infty} (2\pi i d)^{2k} \left(J_2^{2k+2,c}(x) - H_k J_1^{2k+2,c}(x) - S_k K_{2k+2}^c(x) \right). \quad (6.8)$$

The series is uniformly convergent. If a, b are rational, (6.6) makes its left side an explicit integral of finite log-Gamma combinations. If c is rational, (6.5) makes every term on the right a finite Hurwitz order-jet expression at a negative integer.

Proof. At a Fourier mode, divide the logarithm product in (4.9) by $\lambda_{n,a}\lambda_{n,b} = \lambda_{n,c}^2(1 - u^2)$. Multiplication by $(1 - u^2)^{-1}$ yields the coefficient of u^{2k}

$$L_{n,c}^2 - H_k L_{n,c} - S_k.$$

This proves the Fourier coefficient identity in (6.8). The sum of the absolute coefficient contributions, after summing over k , is bounded by a constant times $(1 + \log^2(2 + |n|))/(1 + |n|)^2$. Summing over n proves absolute double convergence and uniform convergence in x . The L^2 convolution has exactly these Fourier coefficients and is continuous; Fourier uniqueness proves equality everywhere. $\square$

6.2 Two half-grid zeta-series evaluations

For a particularly explicit family put $c = \frac{1}{2}$, $|d| < \frac{1}{2}$, $a = \frac{1}{2} - d$, $b = \frac{1}{2} + d$, and $\ell = \log(2\pi)$. Write

$$Z(s) = \zeta(s, \tfrac{1}{2}) = (2^s - 1)\zeta(s), \quad A(s) = \Phi(-1, s, \tfrac{1}{2}) = 2^{-s}(\zeta(s, \tfrac{1}{4}) - \zeta(s, \tfrac{3}{4})).$$

Theorem 6.1 specializes to the ordinary integral identities

$$\begin{aligned} \int_0^1 P_a(t) P_b(1 - t) dt = & -\frac{1}{2\pi^2} \sum_{k \geq 0} d^{2k} \left[Z''(2k + 2) + (H_k - 2\ell) Z'(2k + 2) \right. \\ & \left. + (\ell^2 - H_k \ell - S_k - \frac{\pi^2}{4}) Z(2k + 2) \right], \end{aligned} \quad (6.9)$$

$$\begin{aligned} \int_0^1 P_a(t) P_b(\{\tfrac{1}{2} - t\}) dt = & \frac{i}{4\pi} \sum_{k \geq 0} d^{2k} \left[2A'(2k + 2) \right. \\ & \left. + (H_k - 2\ell) A(2k + 2) \right]. \end{aligned} \quad (6.10)$$

For rational a, b , these are explicit log-Gamma integrals in positive integer zeta derivatives. They are new derived instances in this continuation, not assertions of first occurrence in the literature.

To verify the formulas, pair the modes n and $-n - 1$:

$$\begin{aligned} K_s^{1/2}(0) &= 2 \cos(\pi s/2) (2\pi)^{-s} Z(s), \\ K_s^{1/2}(\tfrac{1}{2}) &= -2i \sin(\pi s/2) (2\pi)^{-s} A(s). \end{aligned}$$

Differentiate these identities in s once with a minus sign and twice with a plus sign, and evaluate at $s = 2k + 2$. The second derivative of the cosine contributes the $-\pi^2 Z/4$ term in (6.9). This term would be lost by specializing the phase before differentiating.

7 The unilateral multishift Lerch family and its primitives

For positive shifts $a_1, \dots, a_d$, define initially

$$\mathcal{L}(z; \boldsymbol{\alpha}, \mathbf{a}) = \sum_{n=0}^{\infty} z^n \prod_j (n + a_j)^{-\alpha_j}. \quad (7.1)$$

For $|z| < 1$ it is entire in the orders. At $z = 1$ the defining series converges absolutely when $\Re A > 1$. Choose $c > 0$ and $d_j = a_j - c$ with $R = \max |d_j| < c$; the midpoint of the extreme positive shifts again works.

Theorem 7.1 (Single-Lerch expansion and meromorphic zeta continuation). *For $|z| < 1$,*

$$\mathcal{L}(z; \boldsymbol{\alpha}, \mathbf{a}) = \sum_{k \geq 0} (-1)^k h_k(\boldsymbol{\alpha}; \mathbf{d}) \Phi(z, A + k, c). \quad (7.2)$$

This is normally convergent on compact order sets and compact subsets of the disk. At $z = 1$ it gives the meromorphic continuation

$$\mathcal{Z}(\boldsymbol{\alpha}; \mathbf{a}) = \sum_{k \geq 0} (-1)^k h_k(\boldsymbol{\alpha}; \mathbf{d}) \zeta(A + k, c). \quad (7.3)$$

The only possible polar hyperplanes are $A = 1 - k$, $k \geq 0$, with residue in the coordinate $A - (1 - k)$ equal to

$$(-1)^k h_k(\boldsymbol{\alpha}; \mathbf{d}) \Big|_{A=1-k}. \quad (7.4)$$

A vanishing residue removes the corresponding pole; the theorem does not assert that every possible hyperplane is an actual divisor.

Proof. Expand the product in (7.1) by (3.3) with $t = -1/(n + c)$. In the disk, exponential damping justifies the double sum locally uniformly. At $z = 1$, interchange is first justified for $\Re A > 1$. For a compact order set and sufficiently large k , $|\zeta(A + k, c)| \leq Cc^{-k}$; finitely many earlier terms are treated individually. The bound $|h_k| \leq CR_0^k$ for a fixed $R < R_0 < c$ proves normal convergence away from the finitely many hyperplanes meeting the compact set. Only the k th term can have a pole at $A = 1 - k$, and the Hurwitz residue is one. This proves the stated continuation and residue formula. $\square$

Alternatively one may use any positive centre after taking a finite head. Explicitly, for N with $\max_j |a_j - c| < N + c$,

$$\begin{aligned} \mathcal{L}(z; \boldsymbol{\alpha}, \mathbf{a}) &= \sum_{n=0}^{N-1} z^n \prod_j (n + a_j)^{-\alpha_j} \\ &\quad + z^N \sum_{k \geq 0} (-1)^k h_k(\boldsymbol{\alpha}; \mathbf{a} - c) \Phi(z, A + k, N + c). \end{aligned} \quad (7.5)$$

This gives the same continuation and is useful when a preferred arithmetic centre is fixed.

7.1 Differential relations and all ordinary primitives

Termwise operations in the disk give the exact relations

$$(z\partial_z + a_j)\mathcal{L}(z; \alpha, \mathbf{a}) = \mathcal{L}(z; \alpha - \mathbf{e}_j, \mathbf{a}), \quad (7.6)$$

$$\partial_{a_j}\mathcal{L}(z; \alpha, \mathbf{a}) = -\alpha_j\mathcal{L}(z; \alpha + \mathbf{e}_j, \mathbf{a}). \quad (7.7)$$

Consequently, on a chosen branch of z^{a_j} ,

$$\int_0^z w^{a_j-1}\mathcal{L}(w; \alpha, \mathbf{a}) dw = z^{a_j}\mathcal{L}(z; \alpha + \mathbf{e}_j, \mathbf{a}). \quad (7.8)$$

There is also an unweighted formula with no change of variable. For each positive integer p ,

$$\frac{1}{(p-1)!} \int_0^z (z-w)^{p-1}\mathcal{L}(w; \alpha, \mathbf{a}) dw = z^p\mathcal{L}(z; (\alpha, \underbrace{1, \dots, 1}_p); (\mathbf{a}, 1, 2, \dots, p)). \quad (7.9)$$

Indeed, integrating w^n produces $z^{n+p}n!/(n+p)! = z^{n+p} \prod_{j=1}^p (n+j)^{-1}$. All order derivatives commute with these operations on compact subsets of the disk. Thus (7.9) supplies exact ordinary antiderivatives of every logarithmically decorated multishift series. Any repeated shift can be combined by adding its orders.

8 Logarithmic cutoff constants: all indices and exact recursions

For $a, b > 0$ and nonnegative integers m, n , let $M = m + n$ and define

$$C_{m,n}(a, b) = \lim_{N \rightarrow \infty} \left\{ \sum_{j=0}^{N-1} \frac{\log^m(j+a) \log^n(j+b)}{j+a} - \frac{\log^{M+1} N}{M+1} \right\}. \quad (8.1)$$

This is a sharp-cutoff finite part. No Abel or spectral-ray convention is silently substituted.

Theorem 8.1 (A convergent all-index Hurwitz-jet formula). *The limit (8.1) exists and is smooth in positive a, b . If $|b-a| < a$, then*

$$C_{m,n}(a, b) = \gamma_M(a) + (-1)^M \sum_{k=1}^{\infty} (a-b)^k \sum_{q=1}^{\min(n,k)} \binom{n}{q} \frac{q!}{k!} \left[\begin{matrix} k \\ q \end{matrix} \right] \zeta^{(M-q)}(k+1, a). \quad (8.2)$$

The series is locally normally convergent in the stated shift domain. For general positive a, b , choose an integer N with $|b-a| < a+N$ and use the exact finite-head identity

$$C_{m,n}(a, b) = \sum_{j=0}^{N-1} \frac{\log^m(j+a) \log^n(j+b)}{j+a} + C_{m,n}(a+N, b+N). \quad (8.3)$$

In particular, $C_{m,0}(a, b) = \gamma_m(a)$ and $C_{m,n}(a, a) = \gamma_M(a)$.

Proof. Subtract the diagonal summand. For $M \geq 1$, the difference is $O(j^{-2}(1 + \log^{M-1} j))$, locally uniformly with its fixed-order parameter derivatives on compact positive shift sets. For $M = 0$ it is zero. The ordinary Stieltjes cutoff limit then proves existence and smoothness, with the absolutely convergent expression

$$C_{m,n}(a, b) = \gamma_M(a) + \sum_{j \geq 0} \frac{\log^m(j+a) \{\log^n(j+b) - \log^n(j+a)\}}{j+a}. \quad (8.4)$$

Replacing $\log N$ by $\log(N+a)$ in the diagonal subtraction changes the limit by zero.

For the formula, specialize (7.3) to two shifts, with centre a and $d = b - a$:

$$\mathcal{Z}(\alpha, \beta; a, b) = \zeta(\alpha + \beta, a) + \sum_{k \geq 1} \frac{(-d)^k (\beta)_k}{k!} \zeta(\alpha + \beta + k, a). \quad (8.5)$$

Subtract $1/(\alpha + \beta - 1)$. The remainder is holomorphic near $(\alpha, \beta) = (1, 0)$. Its $(-1)^M \partial_\alpha^m \partial_\beta^n$ derivative there equals (8.1): the uniform leading tail is $N^{1-\alpha-\beta}/(\alpha + \beta - 1)$, and the difference between this and the subtracted pole gives exactly $\log^{M+1} N/(M+1)$ after differentiation. The remainder and its jets tend to zero, by the same j^{-2} subtraction as in (8.4).

The first term contributes $\gamma_M(a)$. For $k \geq 1$, the undifferentiated $(\beta)_k$ vanishes at zero, so at least one of the n derivatives must hit this factor. Leibniz's rule and (4.4) give (8.2). Normal convergence from Theorem 7.1 justifies the differentiation. Finally, split the sum defining (8.1) at N . The difference between the logarithmic subtractions for the total cutoff and the translated tail cutoff tends to zero, proving (8.3). $\square$

Two useful subfamilies are

$$C_{m,1}(a, b) = \gamma_{m+1}(a) + (-1)^{m+1} \sum_{k \geq 1} \frac{(a-b)^k}{k} \zeta^{(m)}(k+1, a), \quad (8.6)$$

$$C_{m,2}(a, b) = \gamma_{m+2}(a) + 2(-1)^m \sum_{k \geq 1} \frac{(a-b)^k}{k} \left(\zeta^{(m+1)}(k+1, a) + H_{k-1} \zeta^{(m)}(k+1, a) \right). \quad (8.7)$$

The second logarithmic decoration therefore introduces harmonic coefficients even though the zeta values on the right are all single Hurwitz derivatives.

Theorem 8.2 (Lower-degree divided differences and the collision value). *For $b \neq a$ and $n \geq 1$,*

$$\partial_b C_{m,n}(a, b) = \frac{n}{b-a} (C_{m,n-1}(a, b) - C_{n-1,m}(b, a)). \quad (8.8)$$

The apparent collision singularity is removable, with

$$\partial_b C_{m,n}(a, b)|_{b=a} = n(-1)^{M-1} \zeta^{(M-1)}(2, a). \quad (8.9)$$

Consequently all $C_{m,n}$ are determined recursively in total logarithmic degree by

$$C_{m,n}(a, b) = \gamma_M(a) + n \int_a^b \frac{C_{m,n-1}(a, t) - C_{n-1,m}(t, a)}{t-a} dt. \quad (8.10)$$

Proof. Differentiate (8.4), or the defining cutoff whose subtraction is independent of b . The derivative is the absolutely convergent sum

$$n \sum_{j \geq 0} \frac{\log^m(j+a) \log^{n-1}(j+b)}{(j+a)(j+b)}.$$

Apply $1/(XY) = (1/X - 1/Y)/(Y - X)$. The two sums on the right have identical cutoff divergences, which cancel, yielding (8.8). Setting $b = a$ directly in the convergent differentiated series gives (8.9). Integrating from the diagonal gives (8.10). Both constants on the right have total logarithmic degree $M - 1$, so the recurrence is genuinely lower-degree. $\square$

8.1 A Gamma antiderivative and polygamma collision jets

The first decorated constant has the exact formulas

$$C_{0,1}(a, b) = \gamma_1(a) + \int_a^b \frac{\psi(t) - \psi(a)}{t - a} dt, \quad (8.11)$$

$$\int_a^b C_{0,1}(a, t) dt = (b - a)C_{0,1}(a, b) - \log \Gamma(b) + \log \Gamma(a) + (b - a)\psi(a). \quad (8.12)$$

The first identity is (8.10) with $m = 0, n = 1$; the second follows by integration by parts, since $(t - a)\partial_t C_{0,1}(a, t) = \psi(t) - \psi(a)$. For every $j \geq 1$, direct differentiation also gives

$$\partial_b^j C_{0,1}(a, b) \Big|_{b=a} = (-1)^{j-1} (j-1)! \zeta(j+1, a) = \frac{\psi^{(j)}(a)}{j}. \quad (8.13)$$

Thus this two-shift finite part has a complete polygamma Taylor jet on the diagonal, while its integral in the second shift has an explicit log-Gamma correction.

9 Spectral rays at the zero-order intersection

The meromorphic family $\mathcal{Z}$ is generally not jointly regular at the origin of its order variables. This is not a defect of the continuation: the hyperplane $A = 0$ can pass through that point. Nevertheless each transverse ray has a removable one-variable singularity and a completely explicit jet.

Let $w_1, \dots, w_d \in \mathbb{C}$, $W = \sum_j w_j \neq 0$, and define

$$\mathcal{Z}_w(t) = \mathcal{Z}(t\mathbf{w}; \mathbf{a}), \quad D = \sum_j w_j(a_j - c), \quad H_k(t) = h_k(t\mathbf{w}; \mathbf{a} - c).$$

The logarithms of $n + a_j$ remain real before multiplication by the possibly complex weights. This convention avoids an implicit change of branch for the product.

Theorem 9.1 (Every transverse spectral-ray jet). *The singularity of $\mathcal{Z}_w$ at $t = 0$ is removable, and*

$$\mathcal{Z}_w(0) = \frac{1}{2} - \frac{\sum_j w_j a_j}{W}, \quad (9.1)$$

$$\mathcal{Z}'_w(0) = \sum_j w_j \log \Gamma(a_j) - \frac{W}{2} \log(2\pi). \quad (9.2)$$

For every integer $m \geq 1$,

$$\begin{aligned} \mathcal{Z}_w^{(m)}(0) &= W^m \zeta^{(m)}(0, c) + (-1)^m m D W^{m-1} \gamma_{m-1}(c) \\ &\quad + \sum_{k \geq 2} (-1)^k \sum_{q=1}^{\min(m, k)} \binom{m}{q} H_k^{(q)}(0) W^{m-q} \zeta^{(m-q)}(k, c). \end{aligned} \quad (9.3)$$

The infinite series is normally convergent for the chosen centre.

Proof. The terms $k = 0, 1$ of (7.3) are

$$\zeta(Wt, c) - tD\zeta(1 + Wt, c).$$

The second term has the convergent expansion

$$-\frac{D}{W} - D \sum_{j \geq 0} \frac{(-1)^j}{j!} \gamma_j(c) W^j t^{j+1}. \quad (9.4)$$

For $k \geq 2$, $\zeta(Wt + k, c)$ is holomorphic near zero and $H_k(0) = 0$. Thus the constant term is $\zeta(0, c) - D/W = \frac{1}{2} - \sum_j w_j a_j / W$. Differentiating (9.4) and applying Leibniz's rule in the remaining normally convergent series gives (9.3).

At first order,

$$H'_k(0) = \frac{1}{k} \sum_j w_j (a_j - c)^k \quad (k \geq 1),$$

by (4.4). Therefore the first derivative is

$$W\zeta'(0, c) - D\gamma_0(c) + \sum_j w_j \sum_{k \geq 2} \frac{(-1)^k (a_j - c)^k}{k} \zeta(k, c).$$

The convergent Gamma Taylor formula

$$\log \Gamma(c + d) - \log \Gamma(c) - d\psi(c) = \sum_{k \geq 2} \frac{(-1)^k d^k}{k} \zeta(k, c)$$

now gives (9.2), using $\gamma_0(c) = -\psi(c)$ and $\zeta'(0, c) = \log \Gamma(c) - \frac{1}{2} \log(2\pi)$. $\square$

The first derivative is independent of the auxiliary centre and is linear in the ray weights. It yields the zeta-regularized product identity

$$\exp(-Z'_w(0)) = \frac{(2\pi)^{W/2}}{\prod_j \Gamma(a_j)^{w_j}}. \quad (9.5)$$

This is a Gamma-product consequence of the one-dimensional shifted sequence; it does not assert a general absence of multiplicative anomalies for other spectra.

In contrast, (9.1) depends on the ray whenever the shifts are not all equal. For $(a, b) = (1, 2)$ the rays $(\alpha, \beta) = (t, t)$ and $(2t, t)$ have limits -1 and $-5/6$, respectively. There is therefore no joint value at $(0, 0)$ in this example, even though the first ray derivative has the additive Gamma formula. The hypothesis $W \neq 0$ is essential: a ray with $W = 0$ lies in the possible polar hyperplane and is not covered by the theorem.

10 Cyclotomic coordinates and explicit scalar instances

For $1 \leq r \leq q$, let $\omega = e^{2\pi i/q}$ and choose w with $w^q = z$, $0 < |z| < 1$. A root-of-unity filter gives

$$\Phi(z, s, r/q) = q^{s-1} w^{-r} \sum_{h=0}^{q-1} \omega^{-rh} \text{Li}_s(\omega^h w). \quad (10.1)$$

Indeed the inner sum selects positive integers congruent to r modulo q . Changing the chosen root w permutes the summands and cancels against w^{-r} , so the expression is independent of this choice. The apparent singularity at $z = 0$ is removable. On compact subsets of the disk all order derivatives may be taken termwise; at derivative order m this gives

$$\Phi^{(m)}(z, s, r/q) = q^{s-1} w^{-r} \sum_{h=0}^{q-1} \omega^{-rh} \sum_{j=0}^m \binom{m}{j} (\log q)^{m-j} \text{Li}_s^{(j)}(\omega^h w). \quad (10.2)$$

The derivative on $\text{Li}_s^{(j)}$ is again with respect to s .

At $z = 1$, start in $\Re s > 1$ and then continue. One must keep the $h = 0$ zeta pole before expanding at $s = 1$. With $L = \log q$,

$$\gamma_m(r/q) = (-1)^m \left\{ \frac{L^{m+1}}{m+1} + \sum_{j=0}^m \binom{m}{j} L^{m-j} \left[(-1)^j \gamma_j + \sum_{h=1}^{q-1} \omega^{-rh} \text{Li}_1^{(j)}(\omega^h) \right] \right\}. \quad (10.3)$$

This includes the conductor-log contribution of the pole. It follows by expanding $q^{s-1}\zeta(s)$, not by treating every summand as holomorphic at one.

Equations (7.2), (8.2), and (10.2) express rational-shift mixed logarithmic sums in polylogarithm order derivatives at explicit cyclotomic points. These are convergent expansions, not finite period-basis assertions. For example, (8.6) and the half-shift zeta formula give

$$C_{0,1}(\tfrac{1}{2}, \tfrac{3}{4}) = \gamma_1 - 2\gamma \log 2 - \log^2 2 + \sum_{k \geq 1} \frac{(-1)^{k+1}(2^{k+1} - 1)}{4^k k} \zeta(k+1). \quad (10.4)$$

In the left side the cutoff subtraction is $\frac{1}{2} \log^2 N$. The exact Gamma antiderivative (8.12) then gives

$$\int_{1/2}^{3/4} C_{0,1}(\tfrac{1}{2}, t) dt = \frac{1}{4} C_{0,1}(\tfrac{1}{2}, \tfrac{3}{4}) - \log \Gamma(\tfrac{3}{4}) + \tfrac{1}{2} \log \pi - \frac{\gamma}{4} - \frac{1}{2} \log 2. \quad (10.5)$$

The constant itself has not been proved irreducible to a smaller finite collection of special values.

11 Source audit, reproducibility, and integration boundaries

11.1 What was actually inspected

The targeted reading used the current manuscript README and editorial ledger, together with the complete unequal-twist chapter and the normalization portion of the preceding twisted chapter, pinned to `b1df802799851c62a860570c3ac78ee31b9f5c04`. The incoming directory inventory was read, but the contents of its binary ZIPs were not exhaustively retrieved or audited. A Library search of related earlier reports corroborated the distinction between previous shifted-argument correlation work and the present unequal-frequency-shift problem. No inference of mathematical correctness was made from an archive name or a successful earlier replay.

No new mathematical error was found in the two targeted canonical chapters. In particular, the common-frequency-interval qualification of the beta mixture, the unrestricted integer partial fractions, and the separate status of the finite mixed-jet question are appropriate. This report supplies an extension and a precise functional obstruction, not a retraction of those results.

The ledger opening still describes a count of 439 textual source files, whereas its later milestone and the README report 746. This is an editorial synchronization issue; the full inventory was not independently recounted here. The proposed correction is to make the opening use a single authoritative inventory count or explicitly identify its historical milestone, rather than silently treating both numbers as current.

11.2 Mathematical safeguards for integration

The following distinctions are essential. Neither the $k = 0$ term of the central expansion nor the zero Fourier coefficient may be deleted in the twisted calculus. The coefficient $(0)_k$ must not be set to zero before its order derivatives are taken. The integer partial-fraction theorem is not a holomorphic identity in arbitrary orders and cannot simply be differentiated in those integer labels. Finally, the contact term in (4.8), the phase second derivative in (6.9), and the conductor-log term in (10.3) must be retained.

These are demonstrated hazards, not claims that the audited manuscript made these errors. In the new multishift family, a purported joint value at the zero-order intersection is specifically ruled out by (9.1); a ray prescription must be stated.

11.3 Executed checks and their evidentiary role

The package includes `scripts/verify.py`, with separate `exact`, `spectral`, `scalar`, and `gamma` suites. A supplemental verifier, `scripts/verify_extended.py`, checks the general jet compiler and further analytic cases. The exact checks contain 824 assertions in total, grouped as follows.

Finite algebraic family	Assertions
Pochhammer derivatives, including order zero	390
Multicentre coefficient recurrence	27
Symmetric mixed-log coefficients and odd cancellation	48
Gamma-convolution harmonic coefficients	25
Cutoff-jet Stirling coefficients	225
Mixed monodromy certificates	25
Single-centre mixed-monodromy cancellation	20
Integer partial fractions	16
Deliberately corrupted formulas rejected	3
General mixed-jet compiler against independent finite series	45
Total	824

The numerical suites make 38 spectral comparisons, 28 scalar comparisons, four independent ordinary Gamma-integral comparisons, and 26 supplemental comparisons, for 96 numerical comparisons in total at each of 65 and 85 working decimal digits. The scalar comparison for $C_{m,n}$ uses the convergent difference sum (8.4) with a separately computed Euler–Maclaurin tail and a transformed bounded-interval integral; it does not reuse the Hurwitz expansion of the mixed summand. The ordinary integral tests use the finite log-Gamma representatives and compare them with (6.9)–(6.10). Individual residuals and the actual precision settings are recorded in the JSON reports. The supplemental comparisons cover finite Fourier repairs across frequency intervals, ordinary primitives of orders one through three, the pole-aware Stieltjes root filter, the printed quarter-shift constant, and Cauchy extraction of the first three spectral-ray derivatives.

A direct finite-difference call to the numerical polylogarithm routine near order one was unstable at a root of unity. It was replaced by an Euler-transformed defining series, with 150 extra working digits protecting its finite differences. This computational issue is recorded separately from the mathematical identities. The delivered root-filter diagnostics use that independent convergent evaluator rather than the unstable call.

These checks test signs, finite coefficients, branches, zero modes, normalizations, and concrete analytic instances. They do not replace the proofs, establish a numerical independence theorem, or provide interval-certified error bounds. The general distributional and monodromy arguments are ordinary mathematical proofs, not proof-assistant formalizations.

11.4 Suggested placement

The natural home is immediately after `chapters/07-mixed-twist-identities.tex`, in the Hurwitz/Stieltjes volume, with cross-references to the harmonic-number coefficient and polylogarithm chapters. The package includes a short canonical addendum with its own `utj`: labels, an integration map, a theorem-status ledger, and replay/build instructions. The full report should remain as a provenance source. None of these files has been pushed to the repository.

12 Further research questions

- 1. Arithmetic specializations after functional nonclosure.** For rational a, b, c and rational x , identify combinations of (4.7) or (6.8) that collapse to a finite polylogarithmic or Gamma basket. Theorem 5.3 does not exclude these isolated identities. A first concrete target is a finite evaluation, or a nontrivial finite relation, for the constants in (10.4) and their higher logarithmic analogues.
- 2. Larger finite functional spaces.** Determine the precise closure criterion when arbitrary complex orders, polynomial coefficients in the argument, or finitely many spatial translations are allowed. Each enlargement changes the frequency space. The present theorem allows arbitrary finitely many centres, but only integer order ladders and argument-independent coefficients. An obstruction must be proved anew in the enlarged space.
- 3. Minimal mixed-centre logarithmic bases.** Monodromy provides a support invariant recording which centres carry logarithms. Construct a canonical finite reduction at fixed denominator orders and logarithmic degrees in a basis that retains these supports. The rational partial-fraction step should be separated from the cross-centred logarithms rather than forcing them into a single-centre basket which has just been proved insufficient.
- 4. Higher ordinary Gamma products.** Apply the all-jet formula to three or more first primitives and seek explicit symmetry reductions of their finite harmonic coefficients. The multicentre theorem already supplies a convergent representation; the remaining goal is compact identities, especially at rational centres and selected circular arguments, not another existence proof.
- 5. Weighted spectra and higher Gamma functions.** Replace the summand in $\mathcal{Z}$ by a polynomial multiplicity $P(n) \prod_j (n + a_j)^{-\alpha_j}$. Compute the full ray-normalization and first-derivative correction, comparing it with Barnes-type Gamma functions. The cancellation leading to (9.2) uses the unweighted one-dimensional spectrum and should not be assumed to survive unchanged.
- 6. Approach to an integral frequency shift.** As a twist tends to an integer, isolate the exceptional Fourier mode before taking order derivatives. Combine the finite-head formula (3.8) with the canonical untwisted zero-mode subtraction to obtain a uniform bridge between the unequal-twist and untwisted Stieltjes calculi. The limits and normalization should be specified jointly, rather than inferred from pointwise off-endpoint values.
- 7. Multivariable finite parts at intersecting divisors.** The ray formula computes every transverse one-variable jet, but does not select a canonical multivariable finite-part prescription. Determine which coordinate changes preserve a chosen prescription for (7.3), and express the defects by its explicit residue polynomials. This is distinct from assigning a joint value where the ray limits demonstrably disagree.
- 8. Formal verification and independent analytic review.** The finite coefficient recurrence, Stirling identities, and monodromy finite differences are suitable first targets for proof-assistant formalization. A separate analytic layer should formalize the large-integer uniqueness lemma, Fourier pairing estimates, and termwise differentiation. Interval-certified numerical variants may then supplement, rather than substitute for, these proofs.

Conclusion

Unequal twists exhibit an exact dichotomy. Integer rational multipliers admit finite single-centre partial fractions, while genuine multishift logarithmic decoration generally does not. A convergent central expansion replaces the failed finite closure and gives explicit harmonic coefficients at every order. Its primitives are ordinary Gamma convolution identities; its unilateral counterpart gives polylogarithmic primitives, Hurwitz-jet cutoff constants, and controlled spectral-ray finite parts. The distinction between a finite functional identity, an infinite exact expansion, and an isolated arithmetic value remains essential throughout.

References

- [1] ProveIt Contributors, *Unequal twists through beta mixtures and finite resolvent identities*, canonical manuscript source `Analysis/Polylogarithms/docs/manuscript/chapters/07-mixed-twist-identities.tex`, revision `b1df802799851c62a860570c3ac78ee31b9f5c04`. [Pinned source on GitHub](#).
- [2] ProveIt Contributors, *Nonintegral twists: Lerch jets, contact terms, and the half-twist*, canonical manuscript source `Analysis/Polylogarithms/docs/manuscript/chapters/07-twisted-stieltjes.tex`, revision `b1df802799851c62a860570c3ac78ee31b9f5c04`. Targeted reading: family and normalization portions. [Pinned source on GitHub](#).
- [3] ProveIt Contributors, *Source reconciliation and mathematical corrections*, `Analysis/Polylogarithms/docs/manuscript/EDITORIAL-LEDGER.md`, with the manuscript README; inspected 10 October 2026. [Manuscript directory on GitHub](#).
- [4] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz Zeta Function*. In particular the defining series, shift law, polygamma values, argument derivatives, and the log-Gamma value of $\zeta'(0, a)$. <https://dlmf.nist.gov/25.11>.
- [5] NIST Digital Library of Mathematical Functions, §25.14, *Lerch's Transcendent*. <https://dlmf.nist.gov/25.14>.
- [6] NIST Digital Library of Mathematical Functions, §5.7, *Series Expansions of the Gamma and Psi Functions*. <https://dlmf.nist.gov/5.7>.
- [7] NIST Digital Library of Mathematical Functions, §25.12, *Polylogarithms*. <https://dlmf.nist.gov/25.12>.
- [8] NIST Digital Library of Mathematical Functions, §5.15, *Polygamma Functions*. <https://dlmf.nist.gov/5.15>.
- [9] J. C. Lagarias and W.-C. Winnie Li, *The Lerch Zeta Function III. Polylogarithms and Special Values*, *Research in the Mathematical Sciences* **3** (2016), article 2. <https://doi.org/10.1186/s40687-015-0049-2>.